Claim.

For every $n \in \mathbb{N}_{>0}$, we have $n = 1 \vee n > 1$.

Proof.

$n \geq 1$;

To prove $n = 1 \vee n > 1$, consider the following cases.

Case 1. Suppose $n = 1$.

$n = 1 \vee n > 1$;

Case 2. Suppose $n \neq 1$.

We prove $n > 1$ by contradiction. Suppose $\neg\,(n > 1)$.

$n \leq 1$; $n = 1$;

This contradicts $n \neq 1$.

□